COMSATS University Islamabad

Attock Campus

Department Of Computer Science

Course	***IS LAB***
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Assignment No.	***01***

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Assignment 1

Task 1: Implement the Caesar Cipher (10 Marks)

Write a Python program that encrypts and decrypts a message using the Caesar Cipher

```python
def caesar_encrypt(text, shift):  1 usage
    result = ""
    for char in text:
        # Here we check that if the letter is an alphabet we shift it and if it's a number  or space we ignore it
        if char.isalpha():
            if char.isupper():
                start = ord('A')
            else:
                start = ord('a')

            # Apply shifting formula
            #Encrypt: shift the letter forward in the alphabet by 'shift' places,
            # wrap around using %26 if it goes past 'Z' or 'z'
            new_char = chr((ord(char) - start + shift) % 26 + start)
            result += new_char

        else:
            result += char
    return result
```

```python
    return result
# Function to decrypt the message
def caesar_decrypt(text, shift):  1 usage
    result = ""
    for char in text:
        if char.isalpha():

            if char.isupper():
                start = ord('A')
            else:
                start = ord('a')
            # Decrypt: shift the letter backward in the alphabet by 'shift' places,
            # wrap around using %26 if it goes before 'A' or 'a'
            new_char = chr((ord(char) - start - shift) % 26 + start)
            result += new_char

        else:
            result += char

    return result
```

```python
# Input Values For Encryption
message = input("Enter your message: ")
shift = int(input("Enter shift value: "))

encrypted_text = caesar_encrypt(message, shift)
print("Encrypted Text", encrypted_text)

decrypted_text = caesar_decrypt(encrypted_text, shift)
print("Decrypted Text", decrypted_text)
```

Code Explanation:

if char.isalpha():

if char.isupper():

start = ord('A')

else:

start = ord('a')

Here we loop over each character one by one and check if the letter is an alphabet or it is a number or space and in case of number or space, we ignore it

The if condition is to check if the first letter is upper case of lower case in which we assign correct ascii value based on it

Encryption:

new_char = chr((ord(char) - start + shift) % 26 + start)

Step 1: We Convert the letter to is Ascii value

Step 2: We map the letter to alphabetical value

Step 3: We shift the Letter accordingly

Step 4: if the number exceed 25 we % by 26

<u>Decryption Function:</u>

new_char = chr((ord(char) - start - shift) % 26 + start)

Step 1: We Convert the letter to is Ascii value

Step 2: We map the letter to alphabetical value

Step 3: We shift the Letter accordingly to undo the encryption done before

Step 4: if the number goes below zero we % 26

Step 5: Converts The number back to letter

Main Program:

```
message = input("Enter your message: ")
shift = int(input("Enter shift value: "))
encrypted_text = caesar_encrypt(message, shift)
print("Encrypted Text", encrypted_text)
```

Here we take input from user on what they want to encrypt or decrypt

Then ask how many letters to shift

Then the encryption and decryption function is called

Security Analysis:

Caesar Cipher is a simple encryption method that is using letters in the alphabet.

Its greatest weakness is that its breaking can easily happen due to 25 different shifts.

The attacker is free to brute force all the shifts or he can guess the message by examining the frequency of the letters.

In classrooms, it can be taught to encryption but it cannot be regarded as safe with confidential data.